Who Is in the Frame? Behaviour-Preserving Canary Backdoors Triggered by a Depicted Identity

Anonymous CVPR submission

Abstract

[TODO: v0.2: abstract (~200 words), reusing the AAAI framing with updated VLM numbers.]

1. Introduction

[TODO: v0.2. Hook: a backdoor keyed on who is in the frame. Contributions C1–C5 per OUTLINE.md.]

2. Related work

[TODO: v0.2. Map in OUTLINE.md: vision backdoors [2, 3, 11]; LM backdoors [5, 9]; VLM backdoors [6–8]; preservation [1, 4]; defenses [10, 12].]

3. Threat model and method

[TODO: v0.2.]

4. Evaluation protocol

[TODO: v0.2.]

5. Text-only backdoor

[TODO: v0.2. Table 1 from EVAL_REPORT.md.]

6. Identity-triggered VLM backdoor

[TODO: v0.2. Tables 2–3, Figs. 3–4.]

7. Validation on a physical robot

[TODO: Pending real trials (protocol: robot_validation_protocol.md). No numbers until measured.]

8. What went wrong on the way

[TODO: v0.2. Regime ladder A–J; synthetic-square eval artifact.]

9. Limitations and conclusion

[TODO: v0.2.]

LLM Usage Statement

[TODO: Mandatory for CVPR 2027 (desk reject without it). Greg to confirm wording.]

References

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